Course Code: CSE3712	**Course Name:** Big Data Analytics	
Credits: *2-0-2 (Total – 3 Credits) 16 weeks*	**Contact Hours:** 2-hour Theory & 2 hours lab per week	
Batch: 2023, 5th Sem **Academic Year:** 2025-26	**Semester Duration: 29 July- 29 Nov**	
Course Faculty: Dr. Meenakshi [meenakshi.malik@bmu.edu.in **Office:** 63, IV Floor, E2 Building]	**Course Coordinator:** **Name:** Dr. Meenakshi **Email:** meenakshi.malik@bmu.edu.in **Office:** 63, IV Floor, E2 Building	

Aim of the course: The aim of this course "Big Data Analytics" is to discuss techniques to explain how advanced analytics can be leveraged to create competitive advantage and how the data scientist role and skills differ from those of a traditional business intelligence analyst. Students also learn big data and Hadoop ecosystem in order to perform some industrial projects.

Course Overview and Context: Big Data Analytics is an elective course in CSE undergraduate programs. This course provides a foundation in Big Data Analytics. This course includes basics and advanced methods to big data technology and tools. Students will also learn MapReduce, Spark and Hadoop along with its ecosystem.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcomes Statement	**Mapped Component**	**Associated activities/assessment**
CO1	Understand data acquisition process from heterogeneous sources and visualization techniques.	Employability	Hands-on lab sessions, assignments on Industry case studies.
CO2	Perform data processing, cleaning and demonstrate the role of statistics in the analysis of large of datasets.	Skill Development	Practical assignments, Kaggle datasets, Web Scraping, in-class tool demonstrations.
CO3	Solve industry challenges using analytics to inculcate entrepreneurship skills and to develop data science projects, optimize business decisions, and create competitive advantage with analytics.	Entrepreneurship, Skill Development	Capstone project, use of tools

Topics of the course:

	Competency	No. of Sessions	Course outcomes
C1	Understand the types of Digital Data, Concept of data and the characteristics of data.	3(2 Th+ 1 lab)	CO1

C2	Understanding Big data Analytics: Analytics 1.0, 2.0 and 3.0.	3(2 Th+ 1 lab)	CO1
C3	Understanding Lifecycle of data science and big data projects	3(2 Th+ 1 lab)	CO1
C4	Understanding Hadoop architecture and its ecosystem including variant.	3(2 Th+ 1 lab)	CO2
C5	Illustrate Hadoop Distributed File System (HDFS) and its daemons. Understanding the working of HDFS.	3(2 Th+ 1 lab)	CO2
C6	Understanding Map-Reduce and its daemons	3(2 Th+ 1 lab)	CO2
C7	Analyze Word Count problem and solve it using MapReduce.	3(2 Th+ 1 lab)	CO3
C8	Understanding statistical Analysis.	3(2 Th+ 1 lab)	CO2
C9	Understand the basics of Data Visualization.	4(2 Th+ 2 lab)	CO2
C10	Learning and applying tools to analyze big data.	4(3 Th+ 1 lab)	CO3
C11	Understand NoSQL Databases and comparison between NoSQL and RDBMS.	3(2 Th+ 2 lab)	CO3
C12	Case Study and implications of Big Data in Industry	5(3 Th+ 2 lab)	CO3
C13	Understanding and applying classification and clustering.	4(2 Th+ 2 lab)	CO3
C14	Collaborative Filtering, Association Rule Mining	4(2 Th+ 2 lab)	CO3
	Total Sessions:48		

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	3										1	3			
CO2		3											2			
CO3		2	2	3	1									3		

Learning Resources:

Textbooks:

1. Big Data and Analysis, Seema Acrharya and Subhashini, Wiley, First Ed., 2015.

References:

2. Jason Kolb and Jeremy Kolb, The Big Data Revolution, 2013, CreateSpace Independent Publishing Platform.
3. Hadoop in Action, Chuck lam, Dream tech press, reprint ed., 2016.

i) **E-Learning (and other) Resources:** Other relevant study material, MOOCs (and other e-learning material) will be suggested and discussed during the lecture session(s) as and when deemed suitable.

Experiential Learning:

- Project on Data Analytics using Big data tools.
- Demonstration on Big Data Analytics Tools.
- we aim to provide an experiential learning component that encompasses more than 60% of the curriculum.

Assessment Pattern: The final grade will be based on the marks/ grades obtained in the end semester exam along with other assessments defined in the assessment table given below. Relative grading method defined in the academic regulations of the university will be followed to grade the students. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	Weightage (%)	Evaluation Schedule	Rubrics	CO
Quiz	20%	1st week of October	Topics to be covered will be announced in the class. 10 MCQ questions, each of 2 mark. Mode of quiz will be online.	CO1, CO2
Lab Evaluation	10%	1st week of November	Throughout Semester	CO1, CO2
Project Midsem	30%	1st week of October	**Focus:** Working with real world dataset. Data Collection Procedures (20%) Data Cleaning and Preprocessing (10%)	CO2

Project Endsem	40%	2nd week of November	**Focus:** Working with real world dataset. Visualization (20%) Project Evaluation Report (20%) The project is well-documented, transparent, and provides clear value.	CO3
– All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code. – Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity. – There is a mandatory requirement to upload the project to a public repository on GitHub.				–

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.